Curriculum Vita

George McNinch

Personal Data

- **Web**: http://gmcninch.math.tufts.edu
- **Email**: george.mcninch@tufts.edu
- **Address**

  Department of Mathematics
  Tufts University
  Joyce Cummings Center
  177 College Avenue
  Medford, MA 02155
  USA

- **Phone**: +1 617-627-6210
- **Citizenship**: United States of America

Higher Education

- Ph.D. (Mathematics) 1996. University of Oregon (Eugene, OR). Advisor: Gary M. Seitz
- B.S. (Mathematics, Physics) 1990. Samford University (Birmingham AL).

Academic Positions

- 2013–present. Professor Tufts University.
- 2006–2013. Associate Professor, Tufts University.
- 2004–2006. Assistant Professor, Tufts University.
- 1996-2004. Assistant Professor, University of Notre Dame.
- 1990-1996. Graduate Teaching Fellow, University of Oregon.

Grants and Awards

- 2011-2013. National Security Agency. Principal investigator: H9230-11-1-0164.
- 2008-2010. National Security Agency. Principal investigator: H98230-08-1-0110.
- 2004-2008. National Science Foundation. Principal investigator: DMS04 37482.
- 1999-2002. National Science Foundation. Principal investigator: DMS99 70301.

Service to the Profession

- 2022-03. Co-organizer (with E. Sommers): Special Session on Algebraic Groups, AMS sectional meeting at Tufts University.
- 2020-03. Co-organizer (with E. Sommers): Special Session on Algebraic Groups, AMS sectional meeting at Tufts University – **cancelled**.

- 2020, 2016, 2013. Panelist for the National Science Foundation.

- 2019-04. International member, Ph.D. committee, École Polytechnique Fédérale de Lausanne.
- 2014-11. Co-organizer (with S. Garibaldi): Special Session on Exceptional Groups in Physics, Algebra, and Geometry, AMS sectional meeting, UNC Greensboro.
- 2012-01. Co-organizer (with S. Garibaldi): Special Session on Linear algebraic groups: their arithmetic, geometry, and representations, Joint Mathematics Meeting, AMS Boston MA.

Service to Tufts University

- AY2023-2026 – AS&E Library Committee
- AY2023-2024. School of Engineering Curriculum Committee (Math Dept liaison).
- AY2020-2021, AY2021-2022, AY2022-2023 – AS&E Executive Commmittee.
- AY2022-2023 – AS&E Committee on Committees (ExComm liaison)
- AY2020-2021, AY2021-2022 – AS&E Budget and Priorities committee (ExComm liaison)
- AY2013-2014 – AY2018-2019. A&S Committee on Academic Standing and Honors. *Chair*: AY2014-2015 – AY2016-17.
- AY2014-2015 to AY2018-2019. AS&E Budget and Priorities committee Committee. *Chair*: AY2018-2019.
- AY2015-2016. AS&E Undergraduate Admissions and Financial Aid Committee.
- AY2015-2016. AS&E Residential Strategies Working Groups.
- AY2015-2016. A&S Workload Committee.
- AY2013-2014 and AY2014-2015. AS&E Faculty Research Support Advisory Committee.
- AY2012-2013. AS&E Committee on Committees.
- AY2011-2012. Faculty Research Awards Committee.
- AY2005-2006 – AY2009-2010. AS&E Undergraduate Admissions and Financial Aid Committee *Chair*: AY2009-2010

Thesis students

- MS 2026 – Julia Rosedale, Tufts University.
- MS 2026 – Alex Miller, Tufts University.
- MS 2025 – Amanda Tran, Tufts University.
- MS 2024 – Eric Tao, Tufts University.
- Ph.D. 2021 – Nariel Monteiro, Tufts University.
- Ph.D. 2017 – Seth Rothschild, Tufts University.
- AY2016-2017 – Christian Testa (BS), Senior Honors Thesis, Tufts University.
- Ph.D. 2015 – Alex Babinski, Tufts University.
- MS 2014 – Nathan Scheinmann, l'École Polytechnique Fédérale de Lausanne (EPFL), Switzerland.
- AY2012-2013 – Samir Chowdhury (BS), Senior Honors Thesis, Tufts University.
- Ph.D. 2011 – Ellen Goldstein, Tufts University.
- Ph.D. 2007 – Art Weiss, Tufts University.
- MS 2008 – Emily Harvey, Tufts University.
- MS 2007 – Erica Waite, Tufts University.
- MS 2007 – David Earls, Tufts University.

Service for the Tufts Mathematics Department

- AY2025-2026, Mathematics Curriculum Committee.
- AY2025-2026, Hiring Committee Geometry.
- AY2024-2025, Hiring Committee AI.
- AY2022-2023. Hiring Committee NWF algebra/algebraic geometry/number theory.
- AY2022-2023. Graduate admissions committee.
- AY2020-2021, AY2021-2022. Preparator, tenure case of Robert Lemke Oliver.
- AY2021-2022. Course scheduling assistant.
- AY2020-2021. Hiring Committe, NWAP Geometric Group Theory.

- AY2019-2020 and AY2020-2021. Math Department Undergraduate Director.
- 2014–2019. Associate Department Chair
- AY2019-2020. Hiring Committee, NWAP Algebraic Geometry.
- AY2017-2018. Hiring Committee, NWAP Number Theory.
- AY2016-2017. Hiring Committee, Assistant Professor in Complex Geometry.
- AY2015-2016. Hiring Committee Chair, Full Time Lecturer.
- AY2014-2015. Hiring Committee Chair, Assistant Professor in Number Theory.
- AY2013-2014. Interim Graduate Director
- AY2013-2014. Hiring Committee, NWAP algebraic geometry.
- AY2012-2013. Hiring Committee, NWAP.
- 2008–2010. Curriculum Committee Chair
- AY2006-2007. Hiring Committee, 3-year Assistant Professor in Algebraic Groups.
- 2004–2017. Graduate Committee

Workshops, Conferences, Lectures (2017-)

- 2025-10-26 – Invited speaker, Special Session on Groups and Representation Theory at the Fall AMS Central Sectional Meeting at St. Louis University.
- 2025-05 - 2025-07 – Supervised a REU project at Tufts University on Formalization of mathematics.
- 2024-12-02 – Invited speaker, Séminaire algèbre et géométries at the l'Institut Fourier at the Université Grenoble Alpes (France).
- 2024-11-21 – Invited speaker, Groups, Arithmetic and Algebraic Geometry Seminar EFPL Lausanne.
- 2024-11-19 – Invited participant, Groups and Representations EPFL Lausanne, Switzerland.
- 2024-10-20 – Invited speaker, Special Session on Finite Groups, Their Representations, and Related Structures at the Fall AMS Western Sectional Meeting at U California Riverside.
- 2024-10-18 – Invited speaker, Algebra and Number Theory Seminar, University of California at Santa Cruz.
- 2024-09-09 Invited speaker, U Pennsylvania Algebra Seminar.
- 2023-05 – Invited speaker, Special Session on Methods in Non-Semisimple Representation Categories at the AMS Spring Western sectional meeting at California State University, Fresno.
- 2023-05 – Invited speaker, Algebra and Number Theory Seminar, University of California Santa Cruz.
- 2021-11 – Invited speaker, Special Session on Cohomology, Representation Theory, and Lie Theory at the (virtual) AMS Fall Southeastern sectional meeting.
- 2021-04 – Invited (virtual) participant, Oberwolfach workshop on Algebraic Groups.
- 2020-03 – Invited speaker, Conference on The combinatorics and geometry of Jordan type and commuting varieties, Northeastern University (Boston MA) – *virtual*.
- 2019-03 – Colloquium speaker, Northern Illinois University math department, DeKalb, IL.
- 2018-11 – Invited speaker, Workshop on Algebraic Groups, Lie Algebras, and their Representations, Max Planck Institute for Mathematics, Bonn, Germany.
- 2018-06 – Invited speaker, Southeastern Lie Theory conference, University of Georgia (Athens, GA).
- 2018-05 – Invited speaker, Algebraic Groups and Geometrization of the Langlands Program, Univ Lyon and École Normale Superieure Lyon (Lyon, France).
- 2018-04 – Invited speaker, Combinatorial Aspects of Nilpotent Orbits, AMS sectional meeting, Northeastern University (Boston, MA).
- 2018-04 – Participant, workshop on Representations of Finite and Algebraic Groups Mathematical Sciences Research Institute (Berkeley, CA).
- 2017-09 – Invited speaker, workshop on Pseudo-reductive Groups Newcastle University (Newcastle, UK).
- 2017-09 – Invited speaker, Algebra seminar, University of York (York, UK).
- 2017-09 – Invited speaker, ARTIN (Algebra and Representation Theory in the North), University of Manchester (Manchester, UK).

Formalized mathematics

- 2025-08 – Equivalence of R[X] ⊗[R] M with PolynomialModule R M as R[X]-modules, contribution to Mathlib4. #28602
- 2025-08 – Proposal to formalize the notion of C_1 fields in Mathlib4. #28320

Manuscripts

1. Local-global principles for the existence of Levi factors, Preprint (2026).
   With David Harbater (University of Pennsylvania) and Julia Hartmann (University of Pennsylvania).
   Links: [arXiv]
2. Levi decompositions of linear algebraic groups and non-abelian cohomology, Pacific J. Math 336 (2024), no. 1-2, pp. 379-397.
   Links: [PDF], [DOI], [MathReview], [BibTeX]
3. Nilpotent elements and reductive subgroups over a local field, Algebras and Representation Theory 24 (2021), pp. 1479-1522.
   Links: [PDF], [DOI], [Springer], [AlgRepTheory], [MathReview], [BibTeX]
4. Reductive subgroup schemes of a parahoric group scheme, Transformation Groups 25 (2020), no. 1, pp. 217-249.
   Links: [PDF], [DOI], [TransfGroups], [MathReview], [BibTeX], [**Errata**]
5. Central subalgebras of the centralizer of a nilpotent element, Proceedings of the American Mathematical Society 144 (2016), no. 6, pp. 2383–2397.
   With Donna M. Testerman (École Polytechnique Fédérale de Lausanne).
   Links: [PDF], [DOI], [MathReview], [BibTeX]
6. Linearity for actions on vector groups, Journal of Algebra 397 (2014), pp. 666–688.
   Links: [PDF], [DOI], [MathReview], [BibTeX]
7. Levi factors of the special fiber of a parahoric group scheme and tame ramification, Algebras and Representation Theory 17 (2014), no. 2, pp. 469–479.
   Links: [PDF], [DOI], [MathReview], [BibTeX]
8. On the descent of Levi factors, Archiv der Mathematik 100 (2013), no. 1, pp. 7–24.
   Links: [PDF], [DOI], [MathReview], [BibTeX]
9. Some good-filtration subgroups of simple algebraic groups, Journal of Pure and Applied Algebra 217 (2013), no. 12, pp. 2400–2413.
   With Chuck Hague (PBGC).
   Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX]
10. Levi decompositions of a linear algebraic group, Transformation Groups 15 (2010), no. 4, pp. 937–964.
    Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX], [**Errata**]
11. Nilpotent centralizers and Springer isomorphisms, Journal of Pure and Applied Algebra 213 (2009), no. 7, pp. 1346–1363.
    With Donna M. Testerman (École Polytechnique Fédérale de Lausanne).
    Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX]
12. Nilpotent subalgebras of semisimple Lie algebras, Comptes Rendus Mathématique, Académie des Sciences, Paris 347 (2009), no. 9-10, pp. 477–482.
    With Paul Levy (Lancaster University) and Donna M. Testerman (École Polytechnique Fédérale de Lausanne).
    Links: [PDF], [DOI], [MathReview], [BibTeX]
13. The centralizer of a nilpotent section, Nagoya Mathematical Journal 190 (2008), pp. 129–181.
    Links: [PDF], [Euclid], [arXiv], [MathReview], [BibTeX], [**Errata**]
14. Completely reducible Lie subalgebras, Transformation Groups 12 (2007), no. 1, pp. 127–135.
    Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX]
15. Completely reducible SL(2) homomorphisms, Transactions of the American Mathematical Society 359 (2007), no. 9, pp. 4489–4510 (electronic).
    With Donna M. Testerman (École Polytechnique Fédérale de Lausanne).
    Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX]
16. On the centralizer of the sum of commuting nilpotent elements, Journal of Pure and Applied Algebra 206 (2006), no. 1-2, pp. 123–140.
    Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX]
17. Optimal SL(2) homomorphisms, Commentarii Mathematici Helvetici 80 (2005), no. 2, pp. 391–426.
    Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX]
18. Nilpotent orbits over ground fields of good characteristic, Mathematische Annalen 329 (2004), no. 1, pp. 49–85.
    Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX]
19. Sub-principal homomorphisms in positive characteristic, Mathematische Zeitschrift 244 (2003), no. 2, pp. 433–455.
    Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX]
20. Faithful representations of SL(2) over truncated Witt vectors, Journal of Algebra 265 (2003), no. 2, pp. 606–618.

Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX]

21. Adjoint Jordan Blocks, Unpublished (2003).
    Links: [PDF], [arXiv], [BibTeX]
22. Component groups of unipotent centralizers in good characteristic, Journal of Algebra 260 (2003), no. 1, pp. 323–337.
    With Eric Sommers (University of Massachusetts Amherst).
    Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX]
23. The second cohomology of small irreducible modules for simple algebraic groups, Pacific Journal of Mathematics 204 (2002), no. 2, pp. 459–472.
    Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX], [**Errata**]
24. Abelian unipotent subgroups of reductive groups, Journal of Pure and Applied Algebra 167 (2002), no. 2-3, pp. 269–300.
    Links: [PDF], [DOI], [arXiv], [MathReview], [BibTeX], [**Errata**]
25. Filtrations and positive characteristic Howe duality, Mathematische Zeitschrift 235 (2000), no. 4, pp. 651–685.
    Links: [PDF], [DOI], [MathReview], [BibTeX]
26. Semisimplicity of exterior powers of semisimple representations of groups, Journal of Algebra 225 (2000), no. 2, pp. 646–666.
    Links: [PDF], [DOI], [MathReview], [BibTeX], [**Errata**]
27. Semisimple modules for finite groups of Lie type, Journal of the London Mathematical Society 60 (1999), no. 3, pp. 771–792.
    Links: [PDF], [DOI], [MathReview], [BibTeX]
28. Dimensional criteria for semisimplicity of representations, Proceedings of the London Mathematical Society 76 (1998), no. 1, pp. 95–149.
    Links: [PDF], [DOI], [MathReview], [BibTeX], [**Errata**]
29. Semisimplicity in positive characteristic, Algebraic groups and their representations, NATO Adv. Sci. Inst. Ser. C Math. Phys. Sci., no. 517 (1998), pp. 43–52.
    Links: [PDF], [MathReview], [BibTeX]
30. Dimensional criteria for semisimplicity of representations, Ph.D. Thesis, University of Oregon (1996), advisor Gary M. Seitz.
    Links: [Thesis], [MathReview]